



North Carolina State Standards Alignment

For Grades K-8

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About Zearn

Zearn is the 501(c)(3) nonprofit educational organization behind Zearn Math, the top-rated math learning platform used by 1 in 4 elementary-school students and by more than 1 million middle-school students nationwide. Zearn Math supports teachers with research-backed curriculum and digital lessons proven to double student academic growth across all levels of student proficiency. Zearn Math can be used flexibly to support a range of instructional needs including Tier 1 and 2 support, acceleration, intervention, tutoring, after school programming and summer programming.

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Zearn Math is aligned to the North Carolina Standard Course of Study K-8 Mathematics.

We are committed to ensuring that our high-quality instructional materials align to the North Carolina Standard Course of Study K-8 Mathematics to fully support North Carolina's students and teachers. The tables on the following pages, organized by North Carolina Standard Course of Study, identify the alignment between each grade-level standard and Zearn Math lesson(s).

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Kindergarten

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
Counting & Cardinality		
NC.K.CC.1	Know number names and recognize patterns in the counting sequence by: - counting to 100 by ones; - counting to 100 by tens.	Mission 5, Topic C, Lesson 13 Mission 5, Topic D, Lesson 15-18
NC.K.CC.2	Count forward beginning from a given number within the known sequence, instead of having to begin at 1.	Mission 5, Topic C, Lesson 13 Mission 5, Topic D, Lesson 16-18
NC.K.CC.3	Write numbers from 0 to 20. Represent a number of objects with a written numeral 0-20, with 0 representing a count of no objects.	Mission 1, Topic D, Lesson 12-16 Mission 1, Topic E, Lesson 17-22 Mission 1, Topic F, Lesson 24, 26-28 Mission 1, Topic G, Lesson 29-32 Mission 1, Topic H, Lesson 33-37 Mission 4, Topic A, Lesson 1-6 Mission 4, Topic B, Lesson 7-12 Mission 4, Topic E, Lesson 25, 27 Mission 5, Topic A, Lesson 1-4 Mission 5, Topic B, Lesson 6, 9 Mission 5, Topic C, Lesson 10-13
NC.K.CC.4	Understand the relationship between numbers and quantities. - When counting objects, say the number names in the standard order, pairing each object with one and only one number name and each number name with one and only one object (one-to-one correspondence); - Recognize that the last number named tells the number of objects counted regardless of their arrangement (cardinality); - State the number of objects in a group, of up to 5 objects, without counting the objects (perceptual subitizing).	Mission 1, Topic B, Lesson 5-6 Mission 1, Topic C, Lesson 7-11 Mission 1, Topic D, Lesson 12-16 Mission 1, Topic E, Lesson 17-22 Mission 1, Topic F, Lesson 23-28 Mission 1, Topic G, Lesson 29-32 Mission 1, Topic H, Lesson 33-37 Mission 4, Topic H, Lesson 38
NC.K.CC.5	Count to answer “How many?” in the following situations:	Mission 1, Topic C, Lesson 8-11 Mission 1, Topic D, Lesson 12-16 Mission 1, Topic E, Lesson 17-22

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
	- Given a number from 1–20, count out that many objects; - Given up to 20 objects, name the next successive number when an object is added, recognizing the quantity is one more/greater; - Given 20 objects arranged in a line, a rectangular array, and a circle, identify how many; - Given 10 objects in a scattered arrangement, identify how many.	Mission 1, Topic F, Lesson 23-28 Mission 1, Topic G, Lesson 29, 31-32 Mission 1, Topic H, Lesson 33-37 Mission 3, Topic F, Lesson 23-24 Mission 5, Topic A, Lesson 1-5 Mission 5, Topic B, Lesson 6 Mission 5, Topic C, Lesson 10-14 Mission 5, Topic D, Lesson 18-19 Mission 5, Topic E, Lesson 22
NC.K.CC.6	Identify whether the number of objects, within 10, in one group is greater than, less than, or equal to the number of objects in another group, by using matching and counting strategies.	Mission 3, Topic E, Lesson 17-19 Mission 3, Topic F, Lesson 20-24 Mission 3, Topic G, Lesson 25-28 Mission 5, Topic E, Lesson 22
NC.K.CC.7	Compare two numbers, within 10, presented as written numerals.	Mission 3, Topic F, Lesson 20 Mission 3, Topic G, Lesson 25-28
Operations & Algebraic Thinking		
NC.K.OA.1	Represent addition and subtraction, within 10: - Use a variety of representations such as objects, fingers, mental images, drawings, sounds, acting out situations, verbal explanations, or expressions; - Demonstrate understanding of addition and subtraction by making connections among representations.	Mission 4, Topic C, Lesson 13-16 Mission 4, Topic D, Lesson 20-24 Mission 4, Topic F, Lesson 29-30 Mission 4, Topic G, Lesson 33 Mission 4, Topic H, Lesson 38
NC.K.OA.2	Solve addition and subtraction word problems, within 10, using objects or drawings to represent the problem, when solving: - Add to/Take From-Result Unknown - Put Together/ Take Apart (Total Unknown and Two Addends Unknown)	Mission 4, Topic C, Lesson 16-18 Mission 4, Topic D, Lesson 19, 21 Mission 4, Topic F, Lesson 31 Mission 4, Topic G, Lesson 34-36 Mission 4, Topic H, Lesson 37
NC.K.OA.3	Decompose numbers less than or equal to 10 into pairs in more than one way using objects or drawings, and record each decomposition by a drawing or expression.	Mission 1, Topic C, Lesson 9-11 Mission 1, Topic D, Lesson 14-15 Mission 4, Topic A, Lesson 2, 4-6 Mission 4, Topic B, Lesson 7-12 Mission 4, Topic D, Lesson 22-24 Mission 4, Topic E, Lesson 25-28 Mission 4, Topic F, Lesson 32

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
NC.K.OA.4	For any number from 0 to 10, find the number that makes 10 when added to the given number using objects or drawings, and record the answer with a drawing or expression.	Mission 4, Topic H, Lesson 39-40
NC.K.OA.5	Demonstrate fluency with addition and subtraction within 5.	Mission 4, Topic G, Lesson 34-36 Mission 4, Topic H, Lesson 37, 39 Mission 6, Topic A, Lesson 1-2 Mission 6, Topic B, Lesson 5
NC.K.OA.6	Recognize and combine groups with totals up to 5 (conceptual subitizing).	<i>Digital Activities</i> <i>Zearn Math includes digital activities that allow for instantly recognizing a quantity in a structured arrangement.</i>
Numbers & Operations in Base 10		
NC.K.NBT.1	Compose and decompose numbers from 11 to 19 into ten ones and some further ones by: - Using objects or drawings. - Recording each composition or decomposition by a drawing or expression. - Understanding that these numbers are composed of ten ones and one, two, three, four, five, six, seven, eight, or nine ones.	Mission 5, Topic A, Lesson 2-5 Mission 5, Topic B, Lesson 6-9 Mission 5, Topic C, Lesson 10-13 Mission 5, Topic E, Lesson 20-21, 23
Measurement & Data		
NC.K.MD.1	Describe measurable attributes of objects; and describe several different measurable attributes of a single object.	Mission 3, Topic A, Lesson 1-2 Mission 3, Topic C, Lesson 8 Mission 3, Topic D, Lesson 13 Mission 3, Topic E, Lesson 16 Mission 3, Topic H, Lesson 32
NC.K.MD.2	Directly compare two objects with a measurable attribute in common, to see which object has “more of”/“less of” the attribute, and describe the difference.	Mission 3, Topic A, Lesson 1-3 Mission 3, Topic B, Lesson 4-7 Mission 3, Topic C, Lesson 8-12 Mission 3, Topic D, Lesson 13-15 Mission 3, Topic F, Lesson 20 Mission 3, Topic G, Lesson 28 Mission 3, Topic H, Lesson 29-32

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
NC.K.MD.3	Classify objects into given categories; count the numbers of objects in each category and sort the categories by count.	Mission 1, Topic A, Lesson 1-3 Mission 1, Topic B, Lesson 4-6 Mission 1, Topic C, Lesson 7 Mission 2, Topic A, Lesson 1 Mission 2, Topic B, Lesson 7
Geometry		
NC.K.G.1	Describe objects in the environment using names of shapes, and describe the relative positions of objects using positional terms.	Mission 2, Topic A, Lesson 5 Mission 2, Topic B, Lesson 8 Mission 2, Topic C, Lesson 10
NC.K.G.2	Correctly name squares, circles, triangles, rectangles, hexagons, cubes, cones, cylinders, and spheres regardless of their orientations or overall size.	Mission 2, Topic A, Lesson 2-5 Mission 2, Topic B, Lesson 7-8 Mission 2, Topic C, Lesson 9-10
NC.K.G.3	Identify squares, circles, triangles, rectangles, hexagons, cubes, cones, cylinders, and spheres as two-dimensional or three dimensional.	Mission 2, Topic A, Lesson 1 Mission 2, Topic B, Lesson 6 Mission 2, Topic C, Lesson 9-10
NC.K.G.4	Analyze and compare two- and three-dimensional shapes, in different sizes and orientations, using informal language to describe their similarities, differences, attributes and other properties.	Mission 2, Topic A, Lesson 1-4 Mission 2, Topic B, Lesson 6-7 Mission 2, Topic C, Lesson 9-10 Mission 6, Topic A, Lesson 3
NC.K.G.5	Model shapes in the world by: - Building and drawing triangles, rectangles, squares, hexagons, circles. - Building cubes, cones, spheres, and cylinders.	Mission 6, Topic A, Lesson 1-3
NC.K.G.6	Compose larger shapes from simple shapes.	Mission 6, Topic B, Lesson 5-7

1st Grade

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
Operations & Algebraic Thinking		
NC.1.OA.1	Represent and solve addition and subtraction word problems, within 20, with unknowns, by using objects, drawings, and equations with a symbol for the unknown number to represent the problem, when solving: - Add to/Take from-Change Unknown - Put together/Take Apart-Addend Unknown - Compare-Difference Unknown	Mission 1, Topic B, Lesson 4-8 Mission 1, Topic C, Lesson 9-13 Mission 1, Topic G, Lesson 25 Mission 1, Topic H, Lesson 28-32 Mission 2, Topic A, Lesson 3-5, 7-9, 11 Mission 2, Topic B, Lesson 12-13, 17-18, 21 Mission 2, Topic C, Lesson 22-24 Mission 2, Topic D, Lesson 27-29 Mission 3, Topic D, Lesson 10-13 Mission 4, Topic E, Lesson 19-22 Mission 6, Topic A, Lesson 1-2 Mission 6, Topic F, Lesson 25-27
NC.1.OA.2	Represent and solve word problems that call for addition of three whole numbers whose sum is less than or equal to 20, by using objects, drawings, and equations with a symbol for the unknown number.	Mission 2, Topic A, Lesson 1
NC.1.OA.3	Apply the commutative and associative properties as strategies for solving addition problems.	Mission 1, Topic E, Lesson 19-20 Mission 1, Topic F, Lesson 21-24 Mission 2, Topic A, Lesson 2, 6, 10-11
NC.1.OA.4	Solve an unknown-addend problem, within 20, by using addition strategies and/or changing it to a subtraction problem.	Mission 1, Topic G, Lesson 26-27 Mission 1, Topic H, Lesson 29-30, 32 Mission 2, Topic B, Lesson 14-19, 21 Mission 2, Topic C, Lesson 22
NC.1.OA.9	Demonstrate fluency with addition and subtraction within 10.	Mission 1, Topic F, Lesson 21-24 Mission 1, Topic J, Lesson 38-39
NC.1.OA.6	Add and subtract, within 20, using strategies such as: Counting on	Mission 1, Topic C, Lesson 9-13 Mission 1, Topic D, Lesson 14-16

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
	• Making ten • Decomposing a number leading to a ten • Using the relationship between addition and subtraction • Using a number line • Creating equivalent but simpler or known sums	Mission 1, Topic E, Lesson 17-20 Mission 1, Topic G, Lesson 26-27 Mission 1, Topic I, Lesson 33-37 Mission 2, Topic A, Lesson 1-11 Mission 2, Topic B, Lesson 12-21 Mission 2, Topic C, Lesson 25 Mission 2, Topic D, Lesson 27-29
NC.1.OA.7	Apply understanding of the equal sign to determine if equations involving addition and subtraction are true.	Mission 1, Topic E, Lesson 17-19 Mission 2, Topic A, Lesson 6, 10 Mission 2, Topic B, Lesson 20 Mission 2, Topic C, Lesson 25
NC.1.OA.8	Determine the unknown whole number in an addition or subtraction equation involving three whole numbers.	Mission 1, Topic D, Lesson 16 Mission 1, Topic H, Lesson 29-30, 32 Mission 2, Topic A, Lesson 10 Mission 2, Topic B, Lesson 20
Numbers & Operations in Base 10		
NC.1.NBT.1	Count to 150, starting at any number less than 150.	Mission 4, Topic A, Lesson 1-6 Mission 6, Topic B, Lesson 7
NC.1.NBT.2	Understand that the two digits of a two-digit number represent amounts of tens and ones. • Unitize by making a ten from a collection of ten ones. • Model the numbers from 11 to 19 as composed of a ten and one, two, three, four, five, six, seven, eight, or nine ones. • Demonstrate that the numbers 10, 20, 30, 40, 50, 60, 70, 80, 90 refer to one, two, three, four, five, six, seven, eight, or nine tens, with 0 ones.	Mission 2, Topic D, Lesson 26-29 Mission 4, Topic A, Lesson 1-4, 6 Mission 4, Topic F, Lesson 23 Mission 6, Topic B, Lesson 3-4, 8-9
NC.1.NBT.3	Compare two two-digit numbers based on the value of the tens and ones digits, recording the results of comparisons with the symbols $>$, $=$, and $<$.	Mission 4, Topic B, Lesson 7-10 Mission 6, Topic B, Lesson 6
NC.1.NBT.4	Using concrete models or drawings, strategies based on place value, properties of operations, and explaining the reasoning used, add, within 100, in the following situations:	Mission 4, Topic C, Lesson 11-12 Mission 4, Topic D, Lesson 13-18 Mission 4, Topic F, Lesson 23-29 Mission 6, Topic C, Lesson 10-17

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
	- A two-digit number and a one-digit number - A two-digit number and a multiple of 10	Mission 6, Topic D, Lesson 18-19
NC.1.NBT.5	Given a two-digit number, mentally find 10 more or 10 less than the number, without having to count; explain the reasoning used.	Mission 4, Topic A, Lesson 5-6 Mission 6, Topic B, Lesson 5
NC.1.NBT.6	Subtract multiples of 10 in the range 10-90 from multiples of 10 in the range 10-90, explaining the reasoning, using: - Concrete models and drawings - Number lines - Strategies based on place value - Properties of operations - The relationship between addition and subtraction	Mission 4, Topic C, Lesson 11 Mission 6, Topic C, Lesson 10
NC.1.NBT.7	Read and write numerals, and represent a number of objects with a written numeral, to 100.	Mission 6, Topic B, Lesson 4, 7, 9
Measurement & Data		
NC.1.MD.1	Order three objects by length; compare the lengths of two objects indirectly by using a third object.	Mission 3, Topic A, Lesson 1-3 Mission 3, Topic B, Lesson 5-6 Mission 3, Topic C, Lesson 9
NC.1.MD.2	Measure lengths with non-standard units. - Express the length of an object as a whole number of non-standard length units. - Measure by laying multiple copies of a shorter object (the length unit) end to end (iterating) with no gaps or overlaps.	Mission 3, Topic B, Lesson 4-6 Mission 3, Topic C, Lesson 7-9
NC.1.MD.3	Tell and write time in hours and half-hours using analog and digital clocks.	Mission 5, Topic D, Lesson 10-13 Mission 6, Topic E, Lesson 20-24
NC.1.MD.4	Organize, represent, and interpret data with up to three categories. - Ask and answer questions about the total number of data points. - Ask and answer questions about how many in each category.	Mission 3, Topic D, Lesson 10-13

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
	Ask and answer questions about how many more or less are in one category than in another.	Mission 4, Topic A, Lesson 6 Mission 6, Topic E, Lesson 20-21
NC.1.MD.5	Identify quarters, dimes, and nickels and relate their values to pennies.	
Geometry		
NC.1.G.1	Distinguish between defining and non-defining attributes and create shapes with defining attributes by: - Building and drawing triangles, rectangles, squares, trapezoids, hexagons, circles. - Building cubes, rectangular prisms, cones, spheres, and cylinders.	Mission 5, Topic A, Lesson 1-3
NC.1.G.2	Create composite shapes by: - Making a two-dimensional composite shape using rectangles, squares, trapezoids, triangles, and half-circles naming the components of the new shape. - Making a three-dimensional composite shape using cubes, rectangular prisms, cones, and cylinders, naming the components of the new shape.	Mission 5, Topic B, Lesson 4-6
NC.1.G.3	Partition circles and rectangles into two and four equal shares. - Describe the shares as halves and fourths, as half of and fourth of. - Describe the whole as two of, or four of the shares. - Explain that decomposing into more equal shares creates smaller shares.	Mission 5, Topic C, Lesson 7-9 Mission 5, Topic D, Lesson 10

2nd Grade

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
Operations & Algebraic Thinking		
NC.2.OA.1	Represent and solve addition and subtraction word problems, within 100, with unknowns in all positions, by using representations and equations with a symbol for the unknown number to represent the problem, when solving: - One-Step problems: - Add to/Take from-Start Unknown - Compare-Bigger Unknown - Compare-Smaller Unknown - Two-Step problems involving single digits: - Add to/Take from- Change Unknown - Add to/Take From- Result Unknown	Mission 1, Topic B, Lesson 3, 5-6, 8 Mission 4, Topic A, Lesson 1-5 Mission 4, Topic B, Lesson 6-10 Mission 4, Topic C, Lesson 11-14, 16 Mission 4, Topic F, Lesson 31 Mission 7, Topic B, Lesson 7-8, 12-13
NC.2.OA.2	Demonstrate fluency with addition and subtraction, within 20, using mental strategies.	Mission 1, Topic A, Lesson 1-2 Mission 1, Topic B, Lesson 4, 7
NC.2.OA.3	Determine whether a group of objects, within 20, has an odd or even number of members by: - Pairing objects, then counting them by 2s. - Determining whether objects can be placed into two equal groups. - Writing an equation to express an even number as a sum of two equal addends.	Mission 6, Topic D, Lesson 17-19
NC.2.OA.4	Use addition to find the total number of objects arranged in rectangular arrays with up to 5 rows and up to 5 columns; write an equation to express the total as a sum of equal addends	Mission 6, Topic A, Lesson 1-4 Mission 6, Topic B, Lesson 5-9 Mission 6, Topic C, Lesson 10-15
Numbers & Operations in Base 10		
NC.2.NBT.1	Understand that the three digits of a three-digit number represent amounts of hundreds, tens, and ones. Unitize by making a hundred from a collection of ten tens.	Mission 3, Topic A, Lesson 1 Mission 3, Topic B, Lesson 2-3 Mission 3, Topic C, Lesson 4-7 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic E, Lesson 11-15

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
	- Demonstrate that the numbers 100, 200, 300, 400, 500, 600, 700, 800, 900 refer to one, two, three, four, five, six, seven, eight, or nine hundreds, with 0 tens and 0 ones. - Compose and decompose numbers using various groupings of hundreds, tens, and ones.	
NC.2.NBT.2	Count within 1,000; skip-count by 5s, 10s, and 100s.	Mission 3, Topic B, Lesson 2-3 Mission 3, Topic C, Lesson 4 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic E, Lesson 12-15 Mission 3, Topic G, Lesson 19-21 Mission 7, Topic E, Lesson 21 Mission 8, Topic D, Lesson 14
NC.2.NBT.3	Read and write numbers, within 1,000, using base-ten numerals, number names, and expanded form.	Mission 3, Topic C, Lesson 4-7 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic E, Lesson 13-14
NC.2.NBT.4	Compare two three-digit numbers based on the value of the hundreds, tens, and ones digits, using $>$, $=$, and $<$ symbols to record the results of comparisons.	Mission 3, Topic F, Lesson 16-18
NC.2.NBT.5	Demonstrate fluency with addition and subtraction, within 100, by: - Flexibly using strategies based on place value, properties of operations, and/or the relationship between addition and subtraction. - Comparing addition and subtraction strategies, and explaining why they work. - Selecting an appropriate strategy in order to efficiently compute sums and differences.	Mission 1, Topic B, Lesson 3, 5-6, 8 Mission 4, Topic A, Lesson 2-4 Mission 4, Topic B, Lesson 6-8, 10 Mission 4, Topic C, Lesson 11-13 Mission 4, Topic F, Lesson 30 Mission 5, Topic A, Lesson 7 Mission 5, Topic B, Lesson 12 Mission 5, Topic C, Lesson 13-18 Mission 5, Topic D, Lesson 19-20 Mission 7, Topic B, Lesson 7-8, 11-13
NC.2.NBT.6	Add up to three two-digit numbers using strategies based on place value and properties of operations.	Mission 4, Topic D, Lesson 22
NC.2.NBT.7	Add and subtract, within 1,000, relating the strategy to a written method, using: - Concrete models or drawings - Strategies based on place value - Properties of operations - Relationship between addition and subtraction	Mission 4, Topic B, Lesson 9 Mission 4, Topic C, Lesson 14-15 Mission 4, Topic D, Lesson 17-21 Mission 4, Topic E, Lesson 23-28 Mission 4, Topic F, Lesson 29-30 Mission 5, Topic A, Lesson 1-7

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
		Mission 5, Topic B, Lesson 8-12 Mission 5, Topic C, Lesson 13-18 Mission 5, Topic D, Lesson 19-20
NC.2.NBT.8	Mentally add 10 or 100 to a given number 100–900, and mentally subtract 10 or 100 from a given number 100–900.	Mission 3, Topic G, Lesson 19-20 Mission 4, Topic D, Lesson 17 Mission 5, Topic A, Lesson 1-2
Measurement & Data		
NC.2.MD.1	Measure the length of an object in standard units by selecting and using appropriate tools such as rulers, yardsticks, meter sticks, and measuring tapes.	Mission 2, Topic A, Lesson 1-3 Mission 2, Topic B, Lesson 4-5 Mission 2, Topic C, Lesson 6 Mission 7, Topic C, Lesson 14-15 Mission 7, Topic D, Lesson 16-17
NC.2.MD.2	Measure the length of an object twice, using length units of different lengths for the two measurements; describe how the two measurements relate to the size of the unit chosen.	Mission 2, Topic C, Lesson 7 Mission 7, Topic D, Lesson 18
NC.2.MD.3	Estimate lengths in using standard units of inches, feet, yards, centimeters, and meters	Mission 2, Topic B, Lesson 5 Mission 2, Topic D, Lesson 9 Mission 7, Topic D, Lesson 17
NC.2.MD.4	Measure to determine how much longer one object is than another, expressing the length difference in terms of a standard length unit.	Mission 2, Topic C, Lesson 6-7 Mission 2, Topic D, Lesson 9 Mission 7, Topic D, Lesson 19
NC.2.MD.5	Use addition and subtraction, within 100, to solve word problems involving lengths that are given in the same units, using equations with a symbol for the unknown number to represent the problem.	Mission 2, Topic D, Lesson 8-10 Mission 7, Topic E, Lesson 20, 22
NC.2.MD.6	Represent whole numbers as lengths from 0 on a number line diagram with equally spaced points and represent whole-number sums and differences, within 100, on a number line.	Mission 2, Topic D, Lesson 8 Mission 7, Topic A, Lesson 3 Mission 7, Topic E, Lesson 21-22 Mission 7, Topic F, Lesson 24
NC.2.MD.7	Tell and write time from analog and digital clocks to the nearest five minutes, using a.m. and p.m.	Mission 8, Topic D, Lesson 13-15

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
NC.2.MD.8	Solve word problems involving: • Quarters, dimes, nickels, and pennies within 99¢, using ¢ symbols appropriately. • Whole dollar amounts, using the \$ symbol appropriately.	Mission 3, Topic D, Lesson 10 Mission 7, Topic B, Lesson 6-10, 12-13
NC.2.MD.10	Organize, represent, and interpret data with up to four categories. • Draw a picture graph and a bar graph with a single-unit scale to represent a data set. • Solve simple put-together, take-apart, and compare problems using information presented in a picture and a bar graph.	Mission 7, Topic A, Lesson 1-5
Geometry		
NC.2.G.1	Recognize and draw triangles, quadrilaterals, pentagons, and hexagons, having specified attributes; recognize and describe attributes of rectangular prisms and cubes.	Mission 8, Topic A, Lesson 1-5 Mission 8, Topic B, Lesson 6
NC.2.G.3	Partition circles and rectangles into two, three, or four equal shares. • Describe the shares using the words halves, thirds, half of, a third of, fourths, fourth of, quarter of. • Describe the whole as two halves, three thirds, four fourths. • Explain that equal shares of identical wholes need not have the same shape.	Mission 8, Topic B, Lesson 7-8 Mission 8, Topic C, Lesson 9-12 Mission 8, Topic D, Lesson 13

3rd Grade

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
Operations & Algebraic Thinking		
NC.3.OA.1	For products of whole numbers with two factors up to and including 10: - Interpret the factors as representing the number of equal groups and the number of objects in each group. - Illustrate and explain strategies including arrays, repeated addition, decomposing a factor, and applying the commutative and associative properties.	Mission 1, Topic A, Lesson 1-3 Mission 1, Topic B, Lesson 6 Mission 1, Topic C, Lesson 7-10 Mission 1, Topic E, Lesson 15-17 Mission 1, Topic F, Lesson 18-19 Mission 3, Topic A, Lesson 1-2 Mission 3, Topic B, Lesson 4-6 Mission 3, Topic C, Lesson 9-10 Mission 3, Topic D, Lesson 12-14 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 20
NC.3.OA.2	For whole-number quotients of whole numbers with a one-digit divisor and a one-digit quotient: - Interpret the divisor and quotient in a division equation as representing the number of equal groups and the number of objects in each group. - Illustrate and explain strategies including arrays, repeated addition or subtraction, and decomposing a factor.	Mission 1, Topic B, Lesson 4-6 Mission 1, Topic D, Lesson 12-13 Mission 1, Topic E, Lesson 17
NC.3.OA.3	Represent, interpret, and solve one-step problems involving multiplication and division. - Solve multiplication word problems with factors up to and including 10. Represent the problem using arrays, pictures, and/or equations with a symbol for the unknown number to represent the problem. - Solve division word problems with a divisor and quotient up to and including 10. Represent the problem using arrays, pictures, repeated subtraction and/or equations with a symbol for the unknown number to represent the problem.	Mission 1, Topic B, Lesson 4-6 Mission 1, Topic C, Lesson 8-9 Mission 1, Topic D, Lesson 11-13 Mission 1, Topic E, Lesson 14-17 Mission 1, Topic F, Lesson 18 Mission 3, Topic A, Lesson 2-3 Mission 3, Topic B, Lesson 7 Mission 3, Topic C, Lesson 11 Mission 3, Topic D, Lesson 13, 15-16

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
NC.3.OA.6	Solve an unknown-factor problem, by using division strategies and/or changing it to a multiplication problem.	Mission 1, Topic B, Lesson 6 Mission 1, Topic D, Lesson 11 Mission 1, Topic E, Lesson 17 Mission 3, Topic D, Lesson 12-13 Mission 3, Topic E, Lesson 16
NC.3.OA.7	Demonstrate fluency with multiplication and division with factors, quotients and divisors up to and including 10. - Know from memory all products with factors up to and including 10. - Illustrate and explain using the relationship between multiplication and division. - Determine the unknown whole number in a multiplication or division equation relating three whole numbers.	Mission 1, Topic D, Lesson 11-13 Mission 1, Topic E, Lesson 14, 16 Mission 3, Topic B, Lesson 4-5 Mission 3, Topic C, Lesson 10 Mission 3, Topic D, Lesson 12-13 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 19
NC.3.OA.8	Solve two-step word problems using addition, subtraction, and multiplication, representing problems using equations with a symbol for the unknown number.	Mission 1, Topic F, Lesson 20-21 Mission 3, Topic E, Lesson 18 Mission 3, Topic F, Lesson 21 Mission 7, Topic A, Lesson 1-3
NC.3.OA.9	Interpret patterns of multiplication on a hundreds board and/or multiplication table.	Mission 3, Topic A, Lesson 1 Mission 3, Topic D, Lesson 13-14 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 19
Numbers & Operations in Base 10		
NC.3.NBT.2	Add and subtract whole numbers up to and including 1,000. - Use estimation strategies to assess reasonableness of answers. - Model and explain how the relationship between addition and subtraction can be applied to solve addition and subtraction problems. - Use expanded form to decompose numbers and then find sums and differences.	Mission 2, Topic A, Lesson 4-5 Mission 2, Topic B, Lesson 10-11 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18-21
NC.3.NBT.3	Use concrete and pictorial models, based on place value and the properties of operations, to find the product of a one-digit whole number by a multiple of 10 in the range 10–90.	Mission 3, Topic F, Lesson 19-20

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
Numbers & Operations - Fractions		
NC.3.NF.1	Interpret unit fractions with denominators of 2, 3, 4, 6, and 8 as quantities formed when a whole is partitioned into equal parts; • Explain that a unit fraction is one of those parts. • Represent and identify unit fractions using area and length models.	Mission 5, Topic B, Lesson 5-9 Mission 5, Topic C, Lesson 10, 12-13 Mission 5, Topic E, Lesson 20, 22, 24-25, 27 Mission 5, Topic F, Lesson 28-29
NC.3.NF.2	Interpret fractions with denominators of 2, 3, 4, 6, and 8 using area and length models. • Using an area model, explain that the numerator of a fraction represents the number of equal parts of the unit fraction. • Using a number line, explain that the numerator of a fraction represents the number of lengths of the unit fraction from 0.	Mission 5, Topic D, Lesson 14-17, 19 Mission 5, Topic E, Lesson 21, 23, 26
NC.3.NF.3	Represent equivalent fractions with area and length models by: • Composing and decomposing fractions into equivalent fractions using related fractions: halves, fourths and eighths; thirds and sixths. • Explaining that a fraction with the same numerator and denominator equals one whole. • Expressing whole numbers as fractions, and recognize fractions that are equivalent to whole numbers.	Mission 5, Topic C, Lesson 10-11, 13 Mission 5, Topic D, Lesson 16-19 Mission 5, Topic E, Lesson 20-27 Mission 5, Topic F, Lesson 28-29
NC.3.NF.4	Compare two fractions with the same numerator or the same denominator by reasoning about their size, using area and length models, and using the $>$, $<$, and $=$ symbols. Recognize that comparisons are valid only when the two fractions refer to the same whole with denominators: halves, fourths and eighths; thirds and sixths.	Mission 5, Topic C, Lesson 10-11, 13 Mission 5, Topic D, Lesson 18-19 Mission 5, Topic F, Lesson 28-29
Measurement & Data		
NC.3.MD.1	Tell and write time to the nearest minute. Solve word problems involving addition and subtraction of time intervals within the same hour.	Mission 2, Topic A, Lesson 1-5 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18, 21

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
NC.3.MD.2	Solve problems involving customary measurement. - Estimate and measure lengths in customary units to the quarter-inch and half-inch, and feet and yards to the whole unit. - Estimate and measure capacity and weight in customary units to a whole number: cups, pints, quarts, gallons, ounces, and pounds. - Add, subtract, multiply, or divide to solve one-step word problems involving whole number measurements of length, weight, and capacity in the same customary units.	Mission 2, Topic B, Lesson 6-11 Mission 2, Topic C, Lesson 12 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18-21 Mission 6, Topic B, Lesson 5-9 Mission 7, Topic D, Lesson 19, 22 Mission 7, Topic E, Lesson 26
NC.3.MD.3	Represent and interpret scaled picture and bar graphs: - Collect data by asking a question that yields data in up to four categories. - Make a representation of data and interpret data in a frequency table, scaled picture graph, and/or scaled bar graph with axes provided. - Solve one and two-step “how many more” and “how many less” problems using information from these graphs.	Mission 6, Topic A, Lesson 1-4
NC.3.MD.5	Find the area of a rectangle with whole-number side lengths by tiling without gaps or overlaps and counting unit squares.	Mission 4, Topic A, Lesson 1-4 Mission 4, Topic B, Lesson 5-7 Mission 4, Topic D, Lesson 13
NC.3.MD.7	Relate area to the operations of multiplication and addition. - Find the area of a rectangle with whole-number side lengths by tiling it, and show that the area is the same as would be found by multiplying the side lengths. - Multiply side lengths to find areas of rectangles with whole-number side lengths in the context of solving problems, and represent whole-number products as rectangular areas in mathematical reasoning. - Use tiles and/or arrays to illustrate and explain that the area of a rectangle can be found by partitioning it into two smaller rectangles, and that the area of the large rectangle is the sum of the two smaller rectangles.	Mission 4, Topic A, Lesson 4 Mission 4, Topic B, Lesson 5-8 Mission 4, Topic C, Lesson 9-11 Mission 4, Topic D, Lesson 12 Mission 7, Topic E, Lesson 28-30

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
NC.3.MD.8	Solve problems involving perimeters of polygons, including finding the perimeter given the side lengths, and finding an unknown side length.	Mission 7, Topic C, Lesson 10-17 Mission 7, Topic D, Lesson 18-22 Mission 7, Topic E, Lesson 23-30
Geometry		
NC.3.G.1	Reason with two-dimensional shapes and their attributes. - Investigate, describe, and reason about composing triangles and quadrilaterals and decomposing quadrilaterals. - Recognize and draw examples and non-examples of types of quadrilaterals including rhombuses, rectangles, squares, parallelograms, and trapezoids.	Mission 7, Topic B, Lesson 4-9

4th Grade

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
Operations & Algebraic Thinking		
NC.4.OA.1	Interpret a multiplication equation as a comparison. Multiply or divide to solve word problems involving multiplicative comparisons using models and equations with a symbol for the unknown number. Distinguish multiplicative comparison from additive comparison.	Mission 1, Topic A, Lesson 1 Mission 3, Topic A, Lesson 2-3 Mission 3, Topic B, Lesson 5-6 Mission 3, Topic C, Lesson 9-11 Mission 3, Topic D, Lesson 12-13 Mission 3, Topic G, Lesson 26 Mission 5, Topic G, Lesson 39-40 Mission 7, Topic A, Lesson 4 Mission 7, Topic C, Lesson 14
NC.4.OA.3	Solve two-step word problems involving the four operations with whole numbers. • Use estimation strategies to assess reasonableness of answers. • Interpret remainders in word problems. • Represent problems using equations with a letter standing for the unknown quantity.	Mission 1, Topic D, Lesson 12 Mission 1, Topic E, Lesson 16 Mission 1, Topic F, Lesson 17-19 Mission 3, Topic D, Lesson 12-13 Mission 3, Topic E, Lesson 14, 19 Mission 3, Topic G, Lesson 31-32
NC.4.OA.4	Find all factor pairs for whole numbers up to and including 50 to: • Recognize that a whole number is a multiple of each of its factors. • Determine whether a given whole number is a multiple of a given one-digit number. • Determine if the number is prime or composite.	Mission 3, Topic F, Lesson 22-25
NC.4.OA.5	Generate and analyze a number or shape pattern that follows a given rule.	Mission 5, Topic H, Lesson 41
Numbers & Operations in Base 10		
NC.4.NBT.1	Explain that in a multi-digit whole number, a digit in one place represents 10 times as much as it represents in the place to its right, up to 100,000.	Mission 1, Topic A, Lesson 2-3

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
NC.4.NBT.2	Read and write multi-digit whole numbers up to and including 100,000 using numerals, number names, and expanded form.	Mission 1, Topic A, Lesson 3-4 Mission 1, Topic B, Lesson 5-6
NC.4.NBT.7	Compare two multi-digit numbers up to and including 100,000 based on the values of the digits in each place, using $>$, $=$, and $<$ symbols to record the results of comparisons.	Mission 1, Topic B, Lesson 5-6
NC.4.NBT.4	Add and subtract multi-digit whole numbers up to and including 100,000 using the standard algorithm with place value understanding.	Mission 1, Topic D, Lesson 11-12 Mission 1, Topic E, Lesson 13-16 Mission 1, Topic F, Lesson 17-19
NC.4.NBT.5	Multiply a whole number of up to three digits by a one-digit whole number, and multiply up to two two-digit numbers with place value understanding using area models, partial products, and the properties of operations. Use models to make connections and develop the algorithm	Mission 3, Topic B, Lesson 4-6 Mission 3, Topic C, Lesson 7-11 Mission 3, Topic H, Lesson 34-38
NC.4.NBT.6	Find whole-number quotients and remainders with up to three-digit dividends and one-digit divisors with place value understanding using rectangular arrays, area models, repeated subtraction, partial quotients, properties of operations, and/or the relationship between multiplication and division.	Mission 3, Topic E, Lesson 14-21 Mission 3, Topic G, Lesson 26-33
Numbers & Operations - Fractions		
NC.4.NF.1	Explain why a fraction is equivalent to another fraction by using area and length fraction models, with attention to how the number and size of the parts differ even though the two fractions themselves are the same size.	Mission 5, Topic B, Lesson 7-11 Mission 5, Topic C, Lesson 14-15 Mission 5, Topic D, Lesson 21 Mission 5, Topic E, Lesson 27 Mission 6, Topic B, Lesson 5 Mission 6, Topic D, Lesson 12
NC.4.NF.2	Compare two fractions with different numerators and different denominators, using the denominators 2, 3, 4, 5, 6, 8, 10, 12, and 100. Recognize that comparisons are valid only when the two fractions refer to the same whole. Record the results of comparisons with symbols $>$, $=$, or $<$, and justify the conclusions by:	Mission 5, Topic C, Lesson 12-15 Mission 5, Topic E, Lesson 26-27

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
	- Reasoning about their size and using area and length models. - Using benchmark fractions 0, $\frac{1}{2}$, and a whole. - Comparing common numerator or common denominators.	
NC.4.NF.3	Understand and justify decompositions of fractions with denominators of 2, 3, 4, 5, 6, 8, 10, 12, and 100. - Understand addition and subtraction of fractions as joining and separating parts referring to the same whole. - Decompose a fraction into a sum of unit fractions and a sum of fractions with the same denominator in more than one way using area models, length models, and equations. - Add and subtract fractions, including mixed numbers with like denominators, by replacing each mixed number with an equivalent fraction, and/or by using properties of operations and the relationship between addition and subtraction. - Solve word problems involving addition and subtraction of fractions, including mixed numbers by writing equations from a visual representation of the problem.	Mission 5, Topic A, Lesson 1-6 Mission 5, Topic D, Lesson 16-21 Mission 5, Topic E, Lesson 22-25 Mission 5, Topic F, Lesson 29-34
NC.4.NF.4	Apply and extend previous understandings of multiplication to: - Model and explain how fractions can be represented by multiplying a whole number by a unit fraction, using this understanding to multiply a whole number by any fraction less than one. - Solve word problems involving multiplication of a fraction by a whole number.	Mission 5, Topic A, Lesson 3, 5-6 Mission 5, Topic E, Lesson 23-25 Mission 5, Topic G, Lesson 35-40
NC.4.NF.6	Use decimal notation to represent fractions. - Express, model and explain the equivalence between fractions with denominators of 10 and 100. - Use equivalent fractions to add two fractions with denominators of 10 or 100.	Mission 6, Topic A, Lesson 1-3 Mission 6, Topic B, Lesson 4-6 Mission 6, Topic C, Lesson 11 Mission 6, Topic D, Lesson 12-14 Mission 6, Topic E, Lesson 15

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
	Represent tenths and hundredths with models, making connections between fractions and decimals.	
NC.4.NF.7	Compare two decimals to hundredths by reasoning about their size using area and length models, and recording the results of comparisons with the symbols $>$, $=$, or $<$. Recognize that comparisons are valid only when the two decimals refer to the same whole.	Mission 6, Topic C, Lesson 9-11
Measurement & Data		
NC.4.MD.1	Know relative sizes of measurement units. Solve problems involving metric measurement. - Measure to solve problems involving metric units: centimeter, meter, gram, kilogram, Liter, milliliter. - Add, subtract, multiply, and divide to solve one-step word problems involving whole-number measurements of length, mass, and capacity that are given in metric units.	Mission 2, Topic A, Lesson 1-3 Mission 2, Topic B, Lesson 4 Mission 4, Topic A, Lesson 4 Mission 7, Topic A, Lesson 1-3 Mission 7, Topic B, Lesson 6-9 Mission 7, Topic C, Lesson 12-13
NC.4.MD.2	Use multiplicative reasoning to convert metric measurements from a larger unit to a smaller unit using place value understanding, two-column tables, and length models.	Mission 2, Topic A, Lesson 1-3 Mission 2, Topic B, Lesson 4-5
NC.4.MD.8	Solve word problems involving addition and subtraction of time intervals that cross the hour.	Grade 3, Mission 2, Topic A, Lesson 4-5
NC.4.MD.3	Solve problems with area and perimeter. - Find areas of rectilinear figures with known side lengths. - Solve problems involving a fixed area and varying perimeters and a fixed perimeter and varying areas. - Apply the area and perimeter formulas for rectangles in real world and mathematical problems.	Mission 3, Topic A, Lesson 1-3 Mission 7, Topic D, Lesson 15-16
NC.4.MD.4	Represent and interpret data using whole numbers. Collect data by asking a question that yields numerical data.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
	• Make a representation of data and interpret data in a frequency table, scaled bar graph, and/or line plot. • Determine whether a survey question will yield categorical or numerical data.	
NC.4.MD.6	Develop an understanding of angles and angle measurement. • Understand angles as geometric shapes that are formed wherever two rays share a common endpoint, and are measured in degrees. • Measure and sketch angles in whole-number degrees using a protractor. • Solve addition and subtraction problems to find unknown angles on a diagram in real-world and mathematical problems.	Mission 4, Topic A, Lesson 1 Mission 4, Topic B, Lesson 5-8 Mission 4, Topic C, Lesson 9-11
Geometry		
NC.4.G.1	Draw and identify points, lines, line segments, rays, angles, and perpendicular and parallel lines.	Mission 4, Topic A, Lesson 1-4 Mission 4, Topic D, Lesson 13-15
NC.4.G.2	Classify quadrilaterals and triangles based on angle measure, side lengths, and the presence or absence of parallel or perpendicular lines.	Mission 4, Topic D, Lesson 13-15
NC.4.G.3	Recognize symmetry in a two-dimensional figure, and identify and draw lines of symmetry.	Mission 4, Topic D, Lesson 12

5th Grade

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
Operations & Algebraic Thinking		
NC.5.OA.2	Write, explain, and evaluate numerical expressions involving the four operations to solve up to two-step problems. Include expressions involving: • Parentheses, using the order of operations. • Commutative, associative and distributive properties.	Mission 2, Topic A, Lesson 1 Mission 2, Topic B, Lesson 3-5 Mission 4, Topic D, Lesson 10 Mission 4, Topic H, Lesson 32 Mission 4, Topic D, Lesson 10
NC.5.OA.3	Generate two numerical patterns using two given rules. • Identify apparent relationships between corresponding terms. • Form ordered pairs consisting of corresponding terms from the two patterns. • Graph the ordered pairs on a coordinate plane.	Mission 6, Topic B, Lesson 7
Numbers & Operations in Base 10		
NC.5.NBT.1	Explain the patterns in the place value system from one million to the thousandths place. • Explain that in a multi-digit number, a digit in one place represents 10 times as much as it represents in the place to its right and $\frac{1}{10}$ of what it represents in the place to its left. • Explain patterns in products and quotients when numbers are multiplied by 1,000, 100, 10, 0.1, and 0.01 and/or divided by 10 and 100.	Mission 1, Topic A, Lesson 1-4 Mission 2, Topic A, Lesson 1-2 Mission 2, Topic E, Lesson 16 Mission 2, Topic C, Lesson 10-12 Mission 2, Topic D, Lesson 13-14 Mission 2, Topic G, Lesson 24
NC.5.NBT.3	Read, write, and compare decimals to thousandths. • Write decimals using base-ten numerals, number names, and expanded form. • Compare two decimals to thousandths based on the value of the digits in each place, using $>$, $=$, and $<$ symbols to record the results of comparisons.	Mission 1, Topic B, Lesson 5-6 Mission 1, Topic D, Lesson 9-10 Mission 1, Topic E, Lesson 11-12 Mission 1, Topic F, Lesson 13-15

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
NC.5.NBT.5	Demonstrate fluency with the multiplication of two whole numbers up to a three-digit number by a two-digit number using the standard algorithm.	Mission 2, Topic B, Lesson 5-9 Mission 2, Topic C, Lesson 10-12 Mission 2, Topic F, Lesson 20-23
NC.5.NBT.6	Find quotients with remainders when dividing whole numbers with up to four-digit dividends and two-digit divisors using rectangular arrays, area models, repeated subtraction, partial quotients, and/or the relationship between multiplication and division. Use models to make connections and develop the algorithm.	Mission 2, Topic E, Lesson 16-18 Mission 2, Topic F, Lesson 19-23 Mission 2, Topic H, Lesson 28-29 Mission 4, Topic G, Lesson 30-31
NC.5.NBT.7	Compute and solve real-world problems with multi-digit whole numbers and decimal numbers. - Add and subtract decimals to thousandths using models, drawings or strategies based on place value. - Multiply decimals with a product to thousandths using models, drawings, or strategies based on place value. - Divide a whole number by a decimal and divide a decimal by a whole number, using repeated subtraction or area models. Decimals should be limited to hundredths. - Use estimation strategies to assess reasonableness of answers.	Mission 1, Topic D, Lesson 9-10 Mission 1, Topic E, Lesson 11-12 Mission 1, Topic F, Lesson 13-16 Mission 2, Topic C, Lesson 10-12 Mission 2, Topic D, Lesson 13-14 Mission 2, Topic G, Lesson 24-27 Mission 2, Topic H, Lesson 28-29 Mission 4, Topic E, Lesson 17-18 Mission 4, Topic G, Lesson 29-31
Numbers & Operations - Fractions		
NC.5.NF.1	Add and subtract fractions, including mixed numbers, with unlike denominators using related fractions: halves, fourths and eighths; thirds, sixths, and twelfths; fifths, tenths, and hundredths. - Use benchmark fractions and number sense of fractions to estimate mentally and assess the reasonableness of answers. - Solve one- and two-step word problems in context using area and length models to develop the algorithm. Represent the word problem in an equation.	Mission 3, Topic A, Lesson 1-2 Mission 3, Topic B, Lesson 3-7 Mission 3, Topic C, Lesson 8-12 Mission 3, Topic D, Lesson 13-15 Mission 4, Topic D, Lesson 10-12
NC.5.NF.3	Use fractions to model and solve division problems. Interpret a fraction as an equal sharing context, where a quantity is divided into equal parts.	Mission 4, Topic B, Lesson 2-5

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
	- Model and interpret a fraction as the division of the numerator by the denominator. - Solve one-step word problems involving division of whole numbers leading to answers in the form of fractions and mixed numbers, with denominators of 2, 3, 4, 5, 6, 8, 10, and 12, using area, length, and set models or equations.	
NC.5.NF.4	Apply and extend previous understandings of multiplication to multiply a fraction or whole number by a fraction, including mixed numbers. - Use area and length models to multiply two fractions, with the denominators 2, 3, 4. - Explain why multiplying a given number by a fraction greater than 1 results in a product greater than the given number and when multiplying a given number by a fraction less than 1 results in a product smaller than the given number. - Solve one-step word problems involving multiplication of fractions using models to develop the algorithm.	Mission 4, Topic C, Lesson 6-9 Mission 4, Topic E, Lesson 13-18 Mission 4, Topic F, Lesson 22-24 Mission 4, Topic G, Lesson 25-31, 33 Mission 5, Topic C, Lesson 10-15
NC.5.NF.7	Solve one-step word problems involving division of unit fractions by non-zero whole numbers and division of whole numbers by unit fractions using area and length models, and equations to represent the problem.	Mission 4, Topic D
Measurement & Data		
NC.5.MD.1	Given a conversion chart, use multiplicative reasoning to solve one-step conversion problems within a given measurement system.	Mission 1, Topic A, Lesson 4 Mission 2, Topic D, Lesson 13-15 Mission 4, Topic C, Lesson 9 Mission 4, Topic E, Lesson 19-20 Mission 5, Topic B, Lesson 5
NC.5.MD.2	Represent and interpret data. - Collect data by asking a question that yields data that changes over time. - Make and interpret a representation of data using a line graph.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
	Determine whether a survey question will yield categorical or numerical data, or data that changes over time.	
NC.5.MD.4	Recognize volume as an attribute of solid figures and measure volume by counting unit cubes, using cubic centimeters, cubic inches, cubic feet, and improvised units.	Mission 5, Topic A, Lesson 1-3 Mission 5, Topic B, Lesson 5
NC.5.MD.5	Relate volume to the operations of multiplication and addition. - Find the volume of a rectangular prism with whole-number side lengths by packing it with unit cubes, and show that the volume is the same as would be found by multiplying the edge lengths. - Build understanding of the volume formula for rectangular prisms with whole-number edge lengths in the context of solving problems. - Find volume of solid figures with one-digit dimensions composed of two non-overlapping rectangular prisms.	Mission 5, Topic B, Lesson 4-9
Geometry		
NC.5.G.1	Graph points in the first quadrant of a coordinate plane, and identify and interpret the x and y coordinates to solve problems.	Mission 6, Topic C, Lesson 13-17 Mission 6, Topic D, Lesson 16-21
NC.5.G.3	Classify quadrilaterals into categories based on their properties. - Explain that attributes belonging to a category of quadrilaterals also belong to all subcategories of that category. - Classify quadrilaterals in a hierarchy based on properties.	Mission 5, Topic D, Lesson 16-21

6th Grade

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
Ratios & Proportional Relationships		
NC.6.RP.1	Understand the concept of a ratio and use ratio language to: - Describe a ratio as a multiplicative relationship between two quantities. - Model a ratio relationship using a variety of representations.	Mission 2, Topic A, Lesson 1-2 Mission 2, Topic B, Lesson 3-5 Mission 9, Topic A, Lesson 4
NC.6.RP.2	Understand that ratios can be expressed as equivalent unit ratios by finding and interpreting both unit ratios in context.	Mission 2, Topic C, Lesson 10 Mission 3, Topic A, Lesson 1 Mission 3, Topic C, Lesson 5-7 Mission 9, Topic A, Lesson 6
NC.6.RP.3	Use ratio reasoning with equivalent whole-number ratios to solve real-world and mathematical problems by: - Creating and using a table to compare ratios. - Finding missing values in the tables. - Using a unit ratio. - Converting and manipulating measurements using given ratios. - Plotting the pairs of values on the coordinate plane.	Mission 2, Topic C, Lesson 6-10 Mission 2, Topic D, Lesson 11-14 Mission 2, Topic E, Lesson 15-16 Mission 2, Topic F, Lesson 17 Mission 3, Topic B, Lesson 3-4 Mission 3, Topic C, Lesson 5-9 Mission 9, Topic A, Lesson 4-6
NC.6.RP.4	Use ratio reasoning to solve real-world and mathematical problems with percents by: - Understanding and finding a percent of a quantity as a ratio per 100. - Using equivalent ratios, such as benchmark percents (50%, 25%, 10%, 5%, 1%), to determine a part of any given quantity. - Finding the whole, given a part and the percent.	Mission 3, Topic D, Lesson 10-16 Mission 6, Topic E, Lesson 17 Mission 9, Topic A, Lesson 4-5
The Number System		
NC.6.NS.1	Use visual models and common denominators to: Interpret and compute quotients of fractions.	Mission 4, Topic A, Lesson 3 Mission 4, Topic B, Lesson 4-9

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
	Solve real-world and mathematical problems involving division of fractions.	Mission 4, Topic C, Lesson 10-11 Mission 4, Topic D, Lesson 12-14 Mission 4, Topic E, Lesson 16-17
NC.6.NS.2	Fluently divide using long division with a minimum of a four-digit dividend and interpret the quotient and remainder in context.	Mission 5, Topic D, Lesson 9-11
NC.6.NS.3	Apply and extend previous understandings of decimals to develop and fluently use the standard algorithms for addition, subtraction, multiplication and division of decimals.	Mission 5, Topic B, Lesson 2-4 Mission 5, Topic C, Lesson 7-8 Mission 5, Topic D, Lesson 12-13 Mission 5, Topic E, Lesson 14-15 Mission 6, Topic A, Lesson 4 Mission 8, Topic C, Lesson 12 Mission 9, Topic A, Lesson 6
NC.6.NS.4	Understand and use prime factorization and the relationships between factors to: - Find the unique prime factorization for a whole number. - Find the greatest common factor of two whole numbers less than or equal to 100. - Use the greatest common factor and the distributive property to rewrite the sum of two whole numbers, each less than or equal to 100. - Find the least common multiple of two whole numbers less than or equal to 12 to add and subtract fractions with unlike denominators.	Mission 7, Topic D, Lesson 16-18 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
NC.6.NS.5	Understand and use rational numbers to: - Describe quantities having opposite directions or values. - Represent quantities in real-world contexts, explaining the meaning of 0 in each situation. - Understand the absolute value of a rational number as its distance from 0 on the number line to: - Interpret absolute value as magnitude for a positive or negative quantity in a real-world context. - Distinguish comparisons of absolute value from statements about order.	Mission 7, Topic A, Lesson 1, 5-7 Mission 7, Topic C, Lesson 13

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
NC.6.NS.6	Understand rational numbers as points on the number line and as ordered pairs on a coordinate plane. - On a number line: - Recognize opposite signs of numbers as indicating locations on opposite sides of 0 and that the opposite of the opposite of a number is the number itself. - Find and position rational numbers on a horizontal or vertical number line. - On a coordinate plane: - Understand signs of numbers in ordered pairs as indicating locations in quadrants. - Recognize that when two ordered pairs differ only by signs, the locations of the points are related by reflections across one or both axes. - Find and position pairs of rational numbers on a coordinate plane.	Mission 7, Topic A, Lesson 1-2, 4, 7 Mission 7, Topic C, Lesson 11-15
NC.6.NS.7	Understand ordering of rational numbers. - Interpret statements of inequality as statements about the relative position of two numbers on a number line diagram. - Write, interpret, and explain statements of order for rational numbers in real-world contexts.	Mission 7, Topic A, Lesson 3-4 Mission 7, Topic B, Lesson 8-9
NC.6.NS.8	Solve real-world and mathematical problems by graphing points in all four quadrants of the coordinate plane. Include use of coordinates and absolute value to find distances between points with the same first coordinate or the same second coordinate.	Mission 7, Topic C, Lesson 11, 13-15
NC.6.NS.9	Apply and extend previous understandings of addition and subtraction. - Describe situations in which opposite quantities combine to make 0. - Understand $p + q$ as the number located a distance q from p, in the positive or negative direction depending on the sign of q. Show that a number and its additive inverse create a zero pair. - Understand subtraction of integers as adding the additive inverse, $p - q = p + (-q)$. Show that	Grade 7, Mission 5, Topic A, Lesson 1 Grade 7, Mission 5, Topic B, Lesson 2-3, 5-7 Grade 7, Mission 6, Topic D, Lesson 18

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
the distance between two integers on the number line is the absolute value of their difference. Use models to add and subtract integers from -20 to 20 and describe real-world contexts using sums and differences.		
Expressions & Equations		
NC.6.EE.1	Write and evaluate numerical expressions, with and without grouping symbols, involving whole-number exponents.	Mission 1, Topic F, Lesson 17-18 Mission 6, Topic C, Lesson 12-15
NC.6.EE.2	Write, read, and evaluate algebraic expressions. - Write expressions that record operations with numbers and with letters standing for numbers. - Identify parts of an expression using mathematical terms and view one or more of those parts as a single entity. - Evaluate expressions at specific values of their variables using expressions that arise from formulas used in real-world problems.	Mission 1, Topic B, Lesson 5-6 Mission 1, Topic C, Lesson 9-10 Mission 1, Topic F, Lesson 18 Mission 6, Topic B, Lesson 6, 10-11 Mission 6, Topic C, Lesson 14-15
NC.6.EE.3	Apply the properties of operations to generate equivalent expressions without exponents.	Mission 6, Topic B, Lesson 10-11
NC.6.EE.4	Identify when two expressions are equivalent and justify with mathematical reasoning.	Mission 5, Topic D, Lesson 13 Mission 6, Topic B, Lesson 8, 10-11
NC.6.EE.5	Use substitution to determine whether a given number in a specified set makes an equation true.	Mission 6, Topic A, Lesson 2-5 Mission 6, Topic B, Lesson 8 Mission 6, Topic C, Lesson 15
NC.6.EE.6	Use variables to represent numbers and write expressions when solving a real-world or mathematical problem.	Mission 6, Topic A, Lesson 1, 3-5 Mission 6, Topic B, Lesson 6-7 Mission 7, Topic B, Lesson 8, 10
NC.6.EE.7	Solve real-world and mathematical problems by writing and solving equations of the form: $x + p = q$ in which p, q and x are all nonnegative rational numbers; and,	Mission 6, Topic A, Lesson 3-5 Mission 6, Topic B, Lesson 7

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
	$p \cdot x = q$ for cases in which p, q and x are all nonnegative rational numbers.	
NC.6.EE.8	Reason about inequalities by: - Using substitution to determine whether a given number in a specified set makes an inequality true. - Writing an inequality of the form $x > c$ or $x < c$ to represent a constraint or condition in a real-world or mathematical problem. - Recognizing that inequalities of the form $x > c$ or $x < c$ have infinitely many solutions. - Representing solutions of inequalities on number line diagrams.	Mission 6, Topic A, Lesson 2-5 Mission 6, Topic B, Lesson 8 Mission 6, Topic C, Lesson 15 Mission 7, Topic B, Lesson 8-10
NC.6.EE.9	Represent and analyze quantitative relationships by: - Using variables to represent two quantities in a real-world or mathematical context that change in relationship to one another. - Analyze the relationship between quantities in different representations (context, equations, tables, and graphs).	Mission 6, Topic D, Lesson 16-18
Geometry		
NC.6.G.1	Create geometric models to solve real-world and mathematical problems to: - Find the area of triangles by composing into rectangles and decomposing into right triangles. - Find the area of special quadrilaterals and polygons by decomposing into triangles or rectangles.	Mission 1, Topic A, Lesson 2-3 Mission 1, Topic B, Lesson 4-6 Mission 1, Topic C, Lesson 7-10 Mission 1, Topic D, Lesson 11 Mission 1, Topic G, Lesson 19 Mission 4, Topic D, Lesson 14
NC.6.G.2	Apply and extend previous understandings of the volume of a right rectangular prism to find the volume of right rectangular prisms with fractional edge lengths. Apply this understanding to the context of solving real-world and mathematical problems.	Mission 1, Topic E, Lesson 15 Mission 4, Topic D, Lesson 14-15 Mission 4, Topic E, Lesson 17
NC.6.G.3	Use the coordinate plane to solve real-world and mathematical problems by: Drawing polygons in the coordinate plane given coordinates for the vertices.	Mission 7, Topic C, Lesson 15

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
	Using coordinates to find the length of a side joining points with the same first coordinate or the same second coordinate.	
NC.6.G.4	Represent right prisms and right pyramids using nets made up of rectangles and triangles, and use the nets to find the surface area of these figures. Apply these techniques in the context of solving real-world and mathematical problems.	Mission 1, Topic E, Lesson 12-16 Mission 1, Topic F, Lesson 18 Mission 1, Topic G, Lesson 19
Statistics & Probability		
NC.6.SP.1	Recognize a statistical question as one that anticipates variability in the data related to the question and accounts for it in the answers.	Mission 8, Topic A, Lesson 2 Mission 8, Topic B, Lesson 3, 6-7 Mission 8, Topic D, Lesson 17
NC.6.SP.2	Understand that a set of data collected to answer a statistical question has a distribution which can be described by its center, spread, and overall shape.	Mission 8, Topic B, Lesson 4-5, 7-8 Mission 8, Topic C, Lesson 11 Mission 8, Topic E, Lesson 18
NC.6.SP.3	Understand that both a measure of center and a description of variability should be considered when describing a numerical data set. - Determine the measure of center of a data set and understand that it is a single number that summarizes all the values of that data set. - Understand that a mean is a measure of center that represents a balance point or fair share of a data set and can be influenced by the presence of extreme values within the data set. - Understand the median as a measure of center that is the numerical middle of an ordered data set. - Understand that describing the variability of a data set is needed to distinguish between data sets in the same scale, by comparing graphical representations of different data sets in the same scale that have similar measures of center, but different spreads.	Mission 8, Topic B, Lesson 6 Mission 8, Topic C, Lesson 9-11
NC.6.SP.4	Display numerical data in plots on a number line. Use dot plots, histograms, and box plots to represent data.	Mission 8, Topic B, Lesson 3-8 Mission 8, Topic D, Lesson 16-17

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
Compare the attributes of different representations of the same data.		
NC.6.SP.5	Summarize numerical data sets in relation to their context. - Describe the collected data by: - Reporting the number of observations in dot plots and histograms. - Communicating the nature of the attribute under investigation, how it was measured, and the units of measurement. - Analyze center and variability by: - Giving quantitative measures of center, describing variability, and any overall pattern, and noting any striking deviations. - Justifying the appropriate choice of measures of center using the shape of the data distribution.	Mission 8, Topic A, Lesson 2 Mission 8, Topic B, Lesson 3-7 Mission 8, Topic C, Lesson 9-12 Mission 8, Topic D, Lesson 13-17 Mission 8, Topic E, Lesson 18

7th Grade

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
Ratios & Proportional Relationships		
NC.7.RP.1	Compute unit rates associated with ratios of fractions to solve real-world and mathematical problems.	Mission 2, Topic C, Lesson 8 Mission 4, Topic A, Lesson 2-3 Mission 9, Topic B, Lesson 5
NC.7.RP.2	Recognize and represent proportional relationships between quantities. - Understand that a proportion is a relationship of equality between ratios. - Represent proportional relationships using tables and graphs. - Recognize whether ratios are in a proportional relationship using tables and graphs. - Compare two different proportional relationships using tables, graphs, equations, and verbal descriptions. - Identify the unit rate (constant of proportionality) within two quantities in a proportional relationship using tables, graphs, equations, and verbal descriptions. - Create equations and graphs to represent proportional relationships. - Use a graphical representation of a proportional relationship in context to: - Explain the meaning of any point (x, y). - Explain the meaning of $(0, 0)$ and why it is included. - Understand that the y-coordinate of the ordered pair $(1, r)$ corresponds to the unit rate and explain its meaning.	Mission 2, Topic A, Lesson 2-3 Mission 2, Topic B, Lesson 4-6 Mission 2, Topic C, Lesson 7-9 Mission 2, Topic D, Lesson 10-13 Mission 2, Topic E, Lesson 14-15 Mission 3, Topic A, Lesson 1, 3, 5 Mission 3, Topic B, Lesson 7 Mission 4, Topic A, Lesson 3-5 Mission 5, Topic C, Lesson 9, 12 Mission 5, Topic D, Lesson 14 Mission 9, Topic A, Lesson 3 Mission 9, Topic B, Lesson 5
NC.7.RP.3	Use scale factors and unit rates in proportional relationships to solve ratio and percent problems.	Mission 3, Topic A, Lesson 5 Mission 4, Topic A, Lesson 1-3 Mission 4, Topic B, Lesson 6-9 Mission 4, Topic C, Lesson 10-15 Mission 4, Topic D, Lesson 16 Mission 9, Topic A, Lesson 1-4 Mission 9, Topic B, Lesson 6, 8

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
		Mission 9, Topic C, Lesson 13
The Number System		
NC.7.NS.1	Apply and extend previous understandings of addition and subtraction to add and subtract rational numbers, using the properties of operations, and describing real-world contexts using sums and differences.	Mission 5, Topic A, Lesson 1 Mission 5, Topic B, Lesson 2-7 Mission 6, Topic D, Lesson 18 Mission 7, Topic B, Lesson 6
NC.7.NS.2	Apply and extend previous understandings of multiplication and division. - Understand that a rational number is any number that can be written as a quotient of integers with a non-zero divisor. - Apply properties of operations as strategies, including the standard algorithms, to multiply and divide rational numbers and describe the product and quotient in real-world contexts. - Use division and previous understandings of fractions and decimals. - Convert a fraction to a decimal using long division. - Understand that the decimal form of a rational number terminates in 0s or eventually repeats.	Mission 4, Topic A, Lesson 5 Mission 5, Topic A, Lesson 1 Mission 5, Topic C, Lesson 8-11 Mission 8, Topic D, Lesson 16 Mission 9, Topic A, Lesson 4
NC.7.NS.3	Solve real-world and mathematical problems involving numerical expressions with rational numbers using the four operations.	Mission 5, Topic B, Lesson 7 Mission 5, Topic C, Lesson 12 Mission 5, Topic D, Lesson 13-14 Mission 5, Topic E, Lesson 15-16 Mission 5, Topic F, Lesson 17 Mission 9, Topic A, Lesson 3 Mission 9, Topic B, Lesson 6
Expression & Equations		
NC.7.EE.1	Apply properties of operations as strategies to: - Add, subtract, and expand linear expressions with rational coefficients. - Factor linear expression with an integer GCF.	Mission 6, Topic D, Lesson 18-22 Mission 9, Topic B, Lesson 7

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
NC.7.EE.2	Understand that equivalent expressions can reveal real-world and mathematical relationships. Interpret the meaning of the parts of each expression in context.	Mission 6, Topic B, Lesson 12
NC.7.EE.3	Solve multi-step real-world and mathematical problems posed with rational numbers in algebraic expressions. • Apply properties of operations to calculate with positive and negative numbers in any form. • Convert between different forms of a number and equivalent forms of the expression as appropriate.	Mission 3, Topic C, Lesson 11 Mission 5, Topic C, Lesson 12 Mission 5, Topic F, Lesson 17 Mission 6, Topic A, Lesson 2-6 Mission 6, Topic B, Lesson 11-12
NC.7.EE.4	Use variables to represent quantities to solve real-world or mathematical problems. - Construct equations to solve problems by reasoning about the quantities. • Fluently solve multistep equations with the variable on one side, including those generated by word problems. • Compare an algebraic solution to an arithmetic solution, identifying the sequence of the operations used in each approach. • Interpret the solution in context. - Construct inequalities to solve problems by reasoning about the quantities. • Fluently solve multi-step inequalities with the variable on one side, including those generated by word problems. • Compare an algebraic solution process for equations and an algebraic solution process for inequalities. • Graph the solution set of the inequality and interpret in context.	Mission 5, Topic E, Lesson 15-16 Mission 6, Topic A, Lesson 4-5 Mission 6, Topic B, Lesson 7-12 Mission 6, Topic C, Lesson 13-17 Mission 7, Topic A, Lesson 5 Mission 9, Topic A, Lesson 3 Mission 9, Topic B, Lesson 7
Geometry		
NC.7.G.1	Solve problems involving scale drawings of geometric figures by: • Building an understanding that angle measures remain the same and side lengths are proportional.	Mission 1, Topic A, Lesson 1-6 Mission 1, Topic B, Lesson 7-12 Mission 1, Topic C, Lesson 13 Mission 2, Topic A, Lesson 1 Mission 3, Topic B, Lesson 6 Mission 3, Topic C, Lesson 11

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
	- Using a scale factor to compute actual lengths and areas from a scale drawing. - Creating a scale drawing.	Mission 9, Topic A, Lesson 4 Mission 9, Topic C, Lesson 13
NC.7.G.2	Understand the characteristics of angles and side lengths that create a unique triangle, more than one triangle or no triangle. Build triangles from three measures of angles and/or sides.	Mission 3, Topic A, Lesson 2 Mission 7, Topic B, Lesson 6-10 Mission 7, Topic D, Lesson 17
NC.7.G.4	Understand area and circumference of a circle. - Understand the relationships between the radius, diameter, circumference, and area. - Apply the formulas for area and circumference of a circle to solve problems.	Mission 3, Topic A, Lesson 3-5 Mission 3, Topic B, Lesson 7-9 Mission 3, Topic C, Lesson 10-11 Mission 9, Topic A, Lesson 4 Mission 9, Topic C, Lesson 11-12
NC.7.G.5	Use facts about supplementary, complementary, vertical, and adjacent angles in a multi-step problem to write and solve equations for an unknown angle in a figure.	Mission 7, Topic A, Lesson 2-5
NC.7.G.6	Solve real-world and mathematical problems involving: - Area and perimeter of two-dimensional objects composed of triangles, quadrilaterals, and polygons. - Volume and surface area of pyramids, prisms, or three-dimensional objects composed of cubes, pyramids, and right prisms.	Mission 1, Topic A, Lesson 6 Mission 2, Topic C, Lesson 8 Mission 3, Topic B, Lesson 6 Mission 7, Topic C, Lesson 12-16 Mission 7, Topic D, Lesson 17 Mission 9, Topic A, Lesson 4 Mission 9, Topic B, Lesson 5, 9 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
Statistics & Probability		
NC.7.SP.1	Understand that statistics can be used to gain information about a population by: - Recognizing that generalizations about a population from a sample are valid only if the sample is representative of that population. - Using random sampling to produce representative samples to support valid inferences	Mission 8, Topic C, Lesson 12-14 Mission 8, Topic D, Lesson 15 Mission 8, Topic E, Lesson 20

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
NC.7.SP.2	Generate multiple random samples (or simulated samples) of the same size to gauge the variation in estimates or predictions, and use this data to draw inferences about a population with an unknown characteristic of interest.	Mission 8, Topic C, Lesson 13-14 Mission 8, Topic D, Lesson 15-17 Mission 8, Topic E, Lesson 20
NC.7.SP.3	Recognize the role of variability when comparing two populations. - Calculate the measure of variability of a data set and understand that it describes how the values of the data set vary with a single number. - Understand the mean absolute deviation of a data set is a measure of variability that describes the average distance that points within a data set are from the mean of the data set. - Understand that the range describes the spread of the entire data set. - Understand that the interquartile range describes the spread of the middle 50% of the data. - Informally assess the difference between two data sets by examining the overlap and separation between the graphical representations of two data sets.	Mission 8, Topic C, Lesson 11 Mission 8, Topic D, Lesson 18
NC.7.SP.4	Use measures of center and measures of variability for numerical data from random samples to draw comparative inferences about two populations.	Mission 8, Topic D, Lesson 15-16, 18-19 Mission 8, Topic E, Lesson 20 Mission 9, Topic A, Lesson 3
NC.7.SP.5	Understand that the probability of a chance event is a number between 0 and 1 that expresses the likelihood of the event occurring.	Mission 8, Topic A, Lesson 2-6
NC.7.SP.6	Collect data to calculate the experimental probability of a chance event, observing its long-run relative frequency. Use this experimental probability to predict the approximate relative frequency.	Mission 8, Topic A, Lesson 1, 3-6
NC.7.SP.7	Develop a probability model and use it to find probabilities of simple events.	Mission 8, Topic A, Lesson 3-6 Mission 8, Topic C, Lesson 14 Mission 8, Topic E, Lesson 20

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
a. Develop a uniform probability model by assigning equal probability to all outcomes, and use the model to determine probabilities of events. b. Develop a probability model (which may not be uniform) by repeatedly performing a chance process and observing frequencies in the data generated. c. Compare theoretical and experimental probabilities from a model to observed frequencies; if the agreement is not good, explain possible sources of the discrepancy.		
NC.7.SP.8	Determine probabilities of compound events using organized lists, tables, tree diagrams, and simulation. a. Understand that, just as with simple events, the probability of a compound event is the fraction of outcomes in the sample space for which the compound event occurs. b. For an event described in everyday language, identify the outcomes in the sample space which compose the event, when the sample space is represented using organized lists, tables, and tree diagrams. c. Design and use a simulation to generate frequencies for compound events.	Mission 8, Topic B, Lesson 7-10 Mission 8, Topic A, Lesson 6

8th Grade

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
The Number System		
NC.8.NS.1	Understand that every number has a decimal expansion. Building upon the definition of a rational number, know that an irrational number is defined as a non-repeating, non-terminating decimal.	Mission 8, Topic E, Lesson 14-15
NC.8.NS.2	Use rational approximations of irrational numbers to compare the size of irrational numbers and locate them approximately on a number line. Estimate the value of expressions involving: • Square roots and cube roots to the tenths. • π to the hundredths.	Mission 8, Topic A, Lesson 1 Mission 8, Topic B, Lesson 4-5 Mission 8, Topic D, Lesson 12-13 <i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
Expressions & Equations		
NC.8.EE.1	Develop and apply the properties of integer exponents to generate equivalent numerical expressions.	Mission 7, Topic B, Lesson 2-8 Mission 7, Topic C, Lesson 11, 14
NC.8.EE.2	Use square root and cube root symbols to: • Represent solutions to equations of the form $x^2 = p$ and $x^3 = p$, where p is a positive rational number. • Evaluate square roots of perfect squares and cube roots of perfect cubes for positive numbers less than or equal to 400.	Mission 8, Topic B, Lesson 2-5 Mission 8, Topic C, Lesson 10 Mission 8, Topic D, Lesson 12-13
NC.8.EE.3	Use numbers expressed in scientific notation to estimate very large or very small quantities and to express how many times as much one is than the other.	Mission 7, Topic C, Lesson 9-12, 14 Mission 7, Topic D, Lesson 16
NC.8.EE.4	Perform multiplication and division with numbers expressed in scientific notation to solve real-world problems, including problems where both decimal and scientific notation are used.	Mission 7, Topic C, Lesson 10-14 Mission 7, Topic D, Lesson 16

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
NC.8.EE.7	Solve real-world and mathematical problems by writing and solving equations and inequalities in one variable. - Recognize linear equations in one variable as having one solution, infinitely many solutions, or no solutions. - Solve linear equations and inequalities including multi-step equations and inequalities with the same variable on both sides	Mission 4, Topic B, Lesson 3-9
NC.8.EE.8	Analyze and solve a system of two linear equations in two variables in slope-intercept form. - Understand that solutions to a system of two linear equations correspond to the points of intersection of their graphs because the point of intersection satisfies both equations simultaneously. - Solve real-world and mathematical problems leading to systems of linear equations by graphing the equations. Solve simple cases by inspection.	Mission 3, Topic C, Lesson 13-14 Mission 4, Topic B, Lesson 9 Mission 4, Topic C, Lesson 10-15 Mission 4, Topic D, Lesson 16
Functions		
NC.8.F.1	Understand that a function is a rule that assigns to each input exactly one output. - Recognize functions when graphed as the set of ordered pairs consisting of an input and exactly one corresponding output. - Recognize functions given a table of values or a set of ordered pairs.	Mission 5, Topic A, Lesson 1-2 Mission 5, Topic B, Lesson 3-5 Mission 5, Topic E, Lesson 17 Mission 9, Topic A, Lesson 1
NC.8.F.2	Compare properties of two linear functions each represented in a different way (algebraically, graphically, numerically in tables, or by verbal descriptions).	Mission 5, Topic B, Lesson 7 Mission 5, Topic C, Lesson 8
NC.8.F.3	Identify linear functions from tables, equations, and graphs.	Mission 5, Topic C, Lesson 8-10
NC.8.F.4	Analyze functions that model linear relationships. Understand that a linear relationship can be generalized by $y = mx + b$.	Mission 5, Topic B, Lesson 4, 7 Mission 5, Topic C, Lesson 8-10 Mission 5, Topic D, Lesson 11 Mission 5, Topic E, Lesson 18

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
	- Write an equation in slope-intercept form to model a linear relationship by determining the rate of change and the initial value, given at least two (x, y) values or a graph. - Construct a graph of a linear relationship given an equation in slope-intercept form. - Interpret the rate of change and initial value of a linear function in terms of the situation it models, and in terms of the slope and y-intercept of its graph or a table of values.	
NC.8.F.5	Qualitatively analyze the functional relationship between two quantities. - Analyze a graph determining where the function is increasing or decreasing; linear or nonlinear. - Sketch a graph that exhibits the qualitative features of a real-world function.	Mission 5, Topic B, Lesson 5-6 Mission 5, Topic C, Lesson 10
Geometry		
NC.8.G.2	Use transformations to define congruence. - Verify experimentally the properties of rotations, reflections, and translations that create congruent figures. - Understand that a two-dimensional figure is congruent to another if the second can be obtained from the first by a sequence of rotations, reflections, and translations. - Given two congruent figures, describe a sequence that exhibits the congruence between them.	Mission 1, Topic A, Lesson 2-4, 6 Mission 1, Topic B, Lesson 7-10 Mission 1, Topic C, Lesson 11-13 Mission 1, Topic D, Lesson 14-15 Mission 2, Topic B, Lesson 6-7
NC.8.G.3	Describe the effect of dilations about the origin, translations, rotations about the origin in 90 degree increments, and reflections across the x-axis and y-axis on two-dimensional figures using coordinates.	Mission 1, Topic A, Lesson 5-6 Mission 2, Topic A, Lesson 4-5 Mission 2, Topic C, Lesson 12
NC.8.G.4	Use transformations to define similarity. - Verify experimentally the properties of dilations that create similar figures. - Understand that a two-dimensional figure is similar to another if the second can be obtained from the first by a sequence of	Mission 2, Topic B, Lesson 6-7, 9

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
	rotations, reflections, translations, and dilations. Given two similar two-dimensional figures, describe a sequence that exhibits the similarity between them.	
NC.8.G.5	Use informal arguments to analyze angle relationships. - Recognize relationships between interior and exterior angles of a triangle. - Recognize the relationships between the angles created when parallel lines are cut by a transversal. - Recognize the angle-angle criterion for similarity of triangles. - Solve real-world and mathematical problems involving angles.	Mission 1, Topic D, Lesson 14-16 Mission 2, Topic B, Lesson 8 Mission 2, Topic D, Lesson 13 Mission 9, Topic A, Lesson 2
NC.8.G.6	Explain the Pythagorean Theorem and its converse.	Mission 8, Topic C, Lesson 7, 9
NC.8.G.7	Apply the Pythagorean Theorem and its converse to solve real-world and mathematical problems.	Mission 8, Topic C, Lesson 6-8, 10
NC.8.G.8	Apply the Pythagorean Theorem to find the distance between two points in a coordinate system.	Mission 8, Topic C, Lesson 11
NC.8.G.9	Understand how the formulas for the volumes of cones, cylinders, and spheres are related and use the relationship to solve real-world and mathematical problems.	Mission 5, Topic D, Lesson 13-16 Mission 5, Topic E, Lesson 17-21 Mission 5, Topic F, Lesson 22
Statistics & Probability		
NC.8.SP.1	Construct and interpret scatter plots for bivariate measurement data to investigate patterns of association between two quantities. Investigate and describe patterns such as clustering, outliers, positive or negative association, linear association, and nonlinear association.	Mission 6, Topic A, Lesson 1-2 Mission 6, Topic B, Lesson 3-8
NC.8.SP.2	Model the relationship between bivariate quantitative data to:	Mission 6, Topic B, Lesson 4-6, 8

NC STANDARD COURSE OF STUDY MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
	- Informally fit a straight line for a scatter plot that suggests a linear association. - Informally assess the model fit by judging the closeness of the data points to the line.	
NC.8.SP.3	Use the equation of a linear model to solve problems in the context of bivariate quantitative data, interpreting the slope and y-intercept.	Mission 6, Topic B, Lesson 6, 8
NC.8.SP.4	Understand that patterns of association can also be seen in bivariate categorical data by displaying frequencies and relative frequencies in a two-way table. - Construct and interpret a two-way table summarizing data on two categorical variables collected from the same subjects. - Use relative frequencies calculated for rows or columns to describe possible association between the two variables.	Mission 6, Topic C, Lesson 9-10